UNIVERSIDAD NACIONAL MAYOR DE SAN MARCOS

Doctoral Program in Deep Tech, Artificial Intelligence & Emerging Technologies

Research Methods and Scientific Integrity in AI and Advanced Technologies

MINI SYSTEMATIC REVIEW, PRISMA 2020 FLOW DIAGRAM AND GAP ANALYSIS

DELIVERABLE No. 4

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MINI SYSTEMATIC REVIEW, PRISMA 2020 FLOW DIAGRAM AND GAP ANALYSIS

DELIVERABLE No. 4

Research topic: A Human-in-the-Loop Multimodal AAC Prototype in Spanish to Support Independent Functional Communication in Autistic Children Requiring Very Substantial Support

1. Review Focus

Deliverable No. 3 left the Literature Review section of my protocol as a preliminary evidence map built on two 2026 studies. This deliverable replaces that map with a small but documented systematic review, reported with the PRISMA 2020 statement (Page et al., 2021). It has three parts: the search and the ten studies it produced, the flow diagram, and the gap analysis that feeds back into the thesis.

Review question. What does the peer-reviewed literature published between 2021 and 2026 report about AAC interventions and about intelligent or context-aware AAC supports for minimally speaking or nonspeaking autistic children, and what remains unstudied for a Spanish-language, human-in-the-loop, multimodal AAC prototype aimed at independent functional communication?

The question was fixed before searching and framed with PICOC, the variant of PICO presented in Session 4 for reviews where context matters (Table 1).

Table 1

PICOC Framing of the Review Question

Element	Definition in this review
Population	Autistic children and adolescents who are minimally speaking, minimally verbal, or nonspeaking.
Intervention	Aided AAC, including speech-generating devices and AAC that uses artificial intelligence, context awareness, or automated suggestions.
Comparison	Conventional or static AAC, baseline phases, or no comparator (reviews).
Outcome	Functional communication outcomes: frequency and independence of communicative acts, communicative functions, speech production, and assistance required.
Context	Home, school, or therapy settings in any country. Language and country were recorded for every study so that the contextual gap could be assessed.

Note. Author's own elaboration. The outcomes mirror the three primary outcomes defined in the research protocol (Deliverable No. 3, Section 06).

Result in one line. The search identified 385 records in PubMed and Semantic Scholar; 280 remained after duplicates were removed, and 10 studies were included. They confirm that AAC helps this population and that AAC with automated or context-aware features is promising in very small samples. Two of the ten were carried out in Spanish, but with conventional AAC. No study evaluates an intelligent or context-aware AAC system in Spanish, and none was conducted in Peru or elsewhere in Latin America.

2. Search Documentation

2.1 Concept blocks

The search string combines three concept blocks. Synonyms are joined with OR inside each block, and the blocks are joined with AND, so a record has to mention the technology, the population, and the age group at the same time (Table 2).

Table 2

Concept Blocks Used to Build the Search String

Block	Concept	Terms joined with OR	Role
A	AAC technology	"augmentative and alternative communication", AAC, "speech generating device*", "speech-generating device*", "speech-output technolog*"	Captures the AAC stream
B	Population	autis*, "minimally speaking", "minimally verbal", nonspeaking, "non-speaking"	Captures the target users
C	Age group	child*, preschool*, youth, adolescen*	Keeps the corpus pediatric

Note. The asterisk is the truncation wildcard and quotation marks force an exact phrase.

I decided not to add a fourth block for artificial intelligence. Terms such as "context-aware" or "artificial intelligence" would leave out the reviews and the Spanish-language studies that the gap analysis depends on, and the thesis needs both. The intelligent or context-aware angle was coded during data extraction instead (column Stream in Table 6).

2.2 Search strings as executed in each source

Each source has its own syntax, so the same three blocks were translated rather than pasted. The strings below are copied exactly as they were run on 4 October 2026 and can be replicated by anyone.

PubMed (field-tagged string). Free-text terms were restricted to title and abstract with [tiab] and combined with the MeSH heading for autism. The publication date limit is part of the string.

```
("augmentative and alternative communication"[tiab] OR "AAC"[tiab]
  OR "speech generating device*"[tiab] OR "speech-generating device*"[tiab]
  OR "speech-output technolog*"[tiab])
AND (autis*[tiab] OR "Autism Spectrum Disorder"[Mesh]
  OR "minimally speaking"[tiab] OR "minimally verbal"[tiab]
  OR nonspeaking[tiab] OR "non-speaking"[tiab])
AND (child*[tiab] OR preschool*[tiab] OR youth[tiab] OR adolescen*[tiab])
AND ("2021/01/01"[dp] : "2026/12/31"[dp])
```

Semantic Scholar (bulk search syntax). Semantic Scholar does not read AND and OR. Its bulk search endpoint uses + for AND and | for OR, and matches the query against title and abstract. The year range is passed as a separate parameter (year = 2021-2026).

```
("augmentative and alternative communication" | "speech generating device"
  | "speech-generating device" | "speech output technology")
+ (autism | autistic | "minimally verbal" | "minimally speaking"
  | nonspeaking)
+ (child | children | preschool* | youth | adolescen*)
```

Scopus and Web of Science (prepared, not run). I do not have institutional access to these two databases yet. The translated strings are recorded here so that the full doctoral review can extend this search without redesigning it.

```
Scopus:
TITLE-ABS-KEY(("augmentative and alternative communication"
  OR "speech generating device*"
  OR "speech-generating device*" OR "speech-output technolog*")
  AND (autis* OR "minimally speaking" OR "minimally verbal" OR nonspeaking)
  AND (child* OR preschool* OR youth OR adolescen*))
  AND PUBYEAR > 2020 AND PUBYEAR < 2027

Web of Science:
TS=(("augmentative and alternative communication"
  OR "speech generating device*"
  OR "speech-generating device*" OR "speech-output technolog*")
  AND (autis* OR "minimally speaking" OR "minimally verbal" OR nonspeaking)
  AND (child* OR preschool* OR youth OR adolescen*)) AND PY=(2021-2026)
```

2.3 Search log

Table 3

Search Log by Source

Source	Date searched	Fields searched	Limits at retrieval	Records
PubMed (E-utilities)	4 October 2026	Title, abstract, MeSH	Publication date 2021 to 2026	137
Semantic Scholar (bulk search)	4 October 2026	Title, abstract	Year 2021 to 2026	248
Total identified				**385**

Note. Language and publication type were not used as retrieval limits. They were applied as eligibility criteria during screening, so every exclusion can be traced in the flow diagram. No registers were searched. All records returned by each source were exported, not only the first page of results.

3. Eligibility Criteria and Selection Process

The criteria in Table 4 were written before screening and were not changed afterwards.

Table 4

Inclusion and Exclusion Criteria

Code	Inclusion criterion	Code	Exclusion criterion
I1	Peer-reviewed journal article reporting an empirical study or a systematic review.	E1	Protocol, tutorial, conference paper, thesis, preprint, commentary, or other output without empirical results.
I2	Published between 2021 and 2026.	E2	Full text in a language other than English or Spanish.
I3	Full text in English or Spanish.	E3	Adult-only sample, or mixed sample without separable data on children.
I4	Reports a communication outcome of aided AAC, speech-generating devices, or intelligent, context-aware, or adaptive AAC.	E4	No AAC communication outcome for children, as in opinion surveys, assessment tools, literacy instruction, or partner training alone.
I5	Population is autistic children or adolescents who are minimally speaking, minimally verbal, or nonspeaking.	E5	Population outside the review focus, such as other diagnoses or mixed developmental disabilities.
I6	Reports design, sample, and outcomes clearly enough to extract them.	E6	Method or outcomes reported too thinly to extract.

Note. Duplicates were removed before screening and are counted separately in the flow diagram.

Selection process. Selection had two stages. In the first, the title and abstract of each of the 280 unique records were checked against I1 to I5, and records that were clearly off topic were removed. In the second, the 132 records that passed were assessed against all six criteria, using the abstract. Twenty of them had no abstract available in either source and are reported as not retrieved. One exclusion reason was logged for each of the remaining reports. Duplicates were identified by DOI, by PubMed identifier, and by normalized title.

Limit of ten studies. The assignment asks for ten studies, and 64 reports met the criteria. The ten were chosen by a priority order fixed in advance: first, every journal study that tested AAC with automated, context-aware, or artificial intelligence features with autistic children; second, AAC intervention studies carried out in Spanish; third, systematic reviews centered on the target population; and last, one conventional intervention study with outcomes beyond requesting. The other 54 eligible reports are counted in the flow diagram under code E7 and stay in the screening log for the full doctoral review.

Use of an AI assistant. Retrieval, removal of duplicates, and the first pass of both screening stages were carried out with the help of an AI assistant (Claude, Anthropic), which applied the criteria in Table 4 to every record. Following the rule discussed in Session 4, I checked its decisions on a sample of more than 10% of the records and on all ten included studies before accepting them. There was no second human reviewer, so inter-rater agreement (Cohen's kappa) could not be calculated. This is acknowledged as a limitation in Section 7.

Data extraction and appraisal. For each included study I extracted design, sample, context, country, and language from the abstract and the bibliographic record, and I recorded the six appraisal points discussed in Session 4: fit between design and question, sample size, statistical or visual-analysis rigor, reproducibility, bias controls, and generalisability. The appraisal was used to interpret the evidence, not to exclude studies.

4. The Ten Included Studies

The ten studies are coded A1 to A10 in chronological order. Full references are in the reference list, and the access links are in Appendix A. Table 5 identifies each study and Table 6 characterizes it.

Table 5

Studies Included in the Review

ID	Citation	Title
A1	White et al. (2021)	Augmentative and alternative communication and speech production for individuals with ASD: A systematic review
A2	Chavers Edgar et al. (2024)	Effects of an AAC intervention package on socio-communicative behaviors between minimally speaking autistic children and their peers
A3	Holyfield et al. (2024)	Leveraging communication partner speech to automate augmented input for children on the autism spectrum who are minimally verbal
A4	Holyfield et al. (2025)	Preliminary investigation of context-aware AAC with automated just-in-time cloze phrase response options
A5	Therrien et al. (2025)	A systematic review of AAC interventions using speech generating devices for autistic preschoolers
A6	Pak et al. (2025)	EMT en Español Para Autismo: A collaborative communication intervention approach and single case design pilot study
A7	Zurita Díaz and Calleja Reina (2025)	The impact of EC+ as a multimodal support in intervention for children with autism spectrum disorder
A8	Griffen et al. (2026)	An examination of context-aware AAC with automated response options for academic communication from autistic children
A9	Pitt et al. (2026)	Artificial intelligence for enhancing emotion expression in AAC systems: A preliminary study
A10	Privitera et al. (2026)	Interventions to promote communication in minimally verbal autistic children: A systematic review

Note. Titles are shortened where the original uses the full phrase “augmentative and alternative communication” or a long subtitle.

Table 6

Characterization of the Included Studies

ID	Design	Sample or data	Focus	Country, language	Stream
A1	Systematic review	Single-case and group studies of AAC and speech	Whether AAC affects speech production	United States, English	3

ID	Design	Sample or data	Focus	Country, language	Stream
A2	Single-case, multiple probe across behaviors	4 minimally speaking autistic children, 6 to 9 years	Requesting turns, commenting, and answering questions with peers	United States, English	4
A3	Single-case, adapted alternating treatments	3 minimally verbal autistic children	Prototype that turns the partner's speech into automated augmented input, against a standard display	United States, English	1
A4	Single-case comparison of two AAC conditions	3 young autistic children	Context-aware AAC against traditional AAC in a social routine	United States, English	1
A5	Systematic review	20 experimental studies	Speech-generating devices for autistic preschoolers	United States, English	3
A6	Single-case, multiple baseline across behaviors	2 Spanish-speaking Latina caregivers and their autistic children	Caregiver-delivered intervention in Spanish with speech-generating devices	United States, Spanish	2
A7	Quasi-experimental, repeated measures over 16 weeks	18 children with Level 3 autism, 6 to 12 years	EC+ multimodal AAC application with aided augmented input	Spain, Spanish	2
A8	Single-case, ABAB reversal	3 autistic preschool children	Context-aware AAC against traditional AAC in academic responses	United States, English	1
A9	Preliminary empirical study	16 children, 8 to 13 years (12 neurotypical, 4 autistic)	Facial emotion recognition to support emotion expression in AAC	United States, English	1
A10	Systematic review	18 studies	Communication interventions for minimally verbal autistic children	Italy, English	3

Note. Stream 1 = AAC with automated, context-aware, or artificial intelligence features. Stream 2 = AAC intervention in Spanish. Stream 3 = review-level synthesis. Stream 4 = conventional intervention with outcomes beyond requesting. For reviews, the country is that of the corresponding author.

What the corpus says. The three reviews (A1, A5, A10) agree on the baseline: aided AAC, and speech-generating devices in particular, improve requesting and other communicative behaviors, and they do not hold back speech. They also agree on the weak points, which are small samples, thin reporting of maintenance and generalization, and outcomes concentrated on requesting. The intervention study A2 shows the same pattern at close range, with clear effects for two of four children and weaker effects for the others.

The four studies in Stream 1 are the closest precedent for the thesis. Three of them (A3, A4, A8) come from the same research group and compare a prototype with a traditional display in three

children each. Holyfield et al. (2024) report more visual attention and linguistic participation, with variable results; Holyfield et al. (2025) report more linguistic participation, vocal approximations, and visual attention; and Griffen et al. (2026) report more frequent, more accurate, and faster responses. Pitt et al. (2026) add a different use of artificial intelligence, emotion recognition, with accuracy that varied considerably by emotion and by child.

The two studies in Stream 2 show that AAC research in Spanish exists, which I did not expect. Pak et al. (2025) adapted a naturalistic intervention for two Spanish-speaking families in the United States, and the children increased the quantity and diversity of their communication. Zurita Díaz and Calleja Reina (2025) tested EC+, a multimodal application developed at the University of Málaga, with 18 children over 16 weeks and found gains in communication, social interaction, and behavior. Neither system suggests messages from context, and neither study was run in Latin America.

Appraisal. The single-case designs match their questions, but none has more than four child participants, and follow-up, where it is reported, covers only a few weeks. A3, A4, and A8 evaluate research prototypes, and the one in A3 was simulated by a hidden operator (Wizard of Oz), which makes independent replication difficult. A7 has the largest sample but no comparison group, so its gains cannot be attributed to the application alone. Generalisability is the weakest point across the corpus: every study of intelligent AAC was run in English in the United States.

5. PRISMA 2020 Flow Diagram

Figure 1 follows the PRISMA 2020 template for new reviews that search databases and registers only. Table 7 repeats the counts so that each subtraction can be checked.

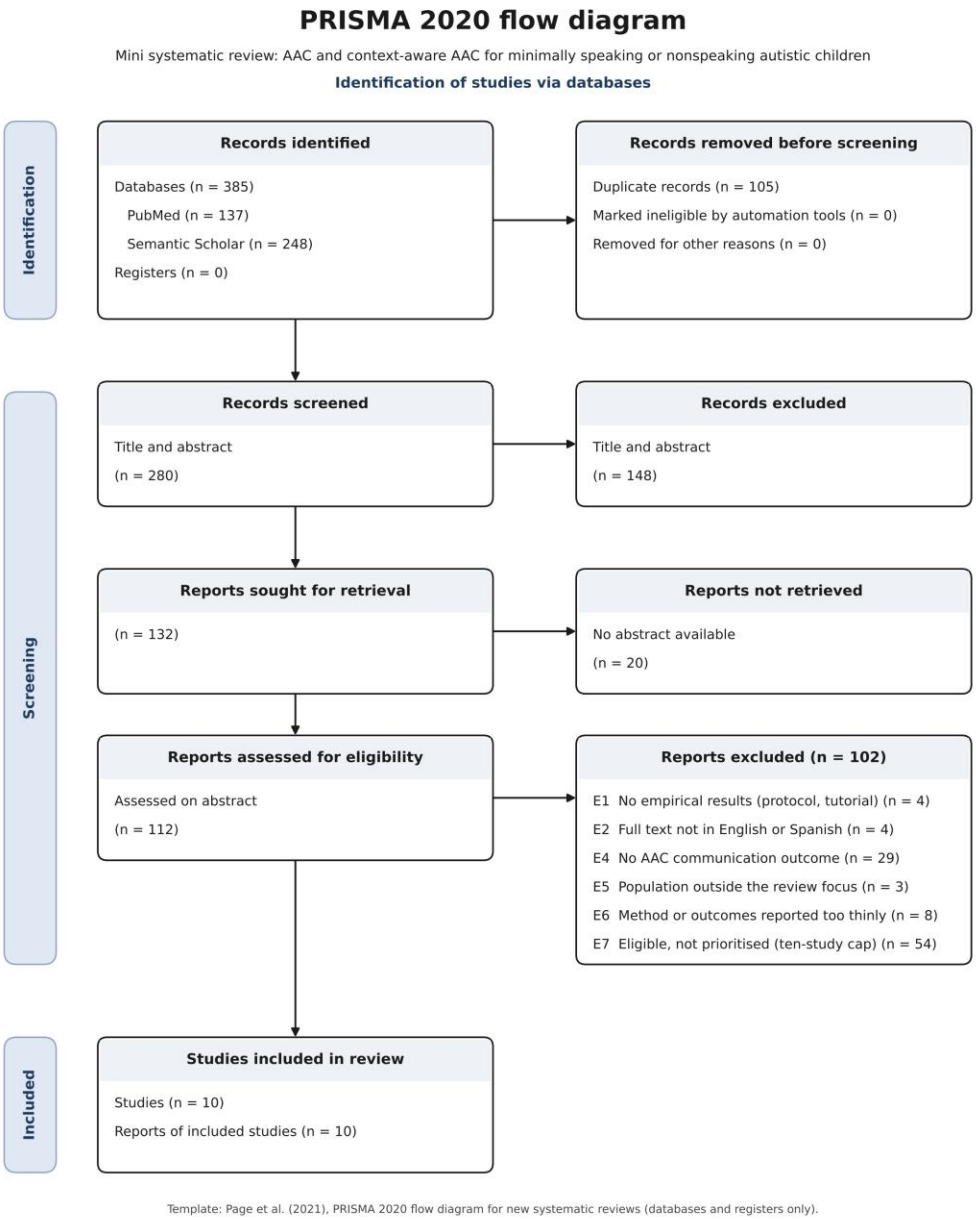


Figure 1

PRISMA 2020 Flow Diagram of the Search and Selection Process

Note. Author's own elaboration based on the PRISMA 2020 flow diagram template (Page et al., 2021).

Table 7

Counts by Phase and Arithmetic Check

Phase	PRISMA box	n	Remaining
Identification	Records from PubMed	137	
Identification	Records from Semantic Scholar	248	385
Identification	Duplicates removed before screening	105	280
Screening	Records excluded on title and abstract	148	132
Screening	Reports not retrieved	20	112
Eligibility	Reports excluded with a reason	102	10
Included	Studies included in the review	10	10

Note. Each value in the last column equals the value above it minus the n of its own row.

Table 8

Reasons for Exclusion at the Eligibility Stage

Code	Reason	n
E1	Protocol, tutorial, or other output without empirical results	4
E2	Full text not in English or Spanish	4
E4	No AAC communication outcome for children	29
E5	Population outside the review focus	3
E6	Method or outcomes reported too thinly to extract	8
E7	Eligible, not prioritised for the ten-study synthesis	54
Total		**102**

Note. One reason was logged per report. No report was excluded under E3 at this stage because adult-only studies had already been removed on title and abstract.

6. Gap Analysis

Table 9 states one candidate gap for each of the four types presented in Session 4, with the studies that support it and the research question of the protocol that responds to it.

Table 9

Candidate Research Gaps Identified in the Corpus

Type of gap	Gap	Evidence in the corpus	Response in the thesis
Knowledge	It is not known whether automated or context-aware suggestions help independent functional communication beyond one narrow task, or what happens when the child can accept, reject, or change a suggestion.	Holyfield et al. (2024, 2025) and Griffen et al. (2026) each test one routine or one academic task. Pitt et al. (2026) test emotion recognition only. No study reports on rejected or modified suggestions.	RQ1 and RQ2. The prototype logs accepted, rejected, and modified suggestions and measures independent acts, communicative diversity, and assistance.
Methodological	Evidence on intelligent AAC rests on three or four participants, short exposure, prototypes that cannot be inspected, and no record of which version of the system was tested.	Holyfield et al. (2024, 2025), Griffen et al. (2026), and Pitt et al. (2026) describe their own studies as preliminary. Therrien et al. (2025) and Privitera et al. (2026) report the same weakness for AAC research in general.	RQ2. Design Science Research with a prespecified protocol, repeated sessions, a double-coded subset, fidelity checks, and versioned artifacts.
Contextual	AAC with automated or context-aware features has not been studied in Spanish, and no included study of any kind was conducted in Peru or elsewhere in Latin America.	The four intelligent AAC studies (A3, A4, A8, A9) were run in English in the United States. The two Spanish-language studies used conventional AAC, with two families in the United States (Pak et al., 2025) and with 18 children in Spain (Zurita Díaz & Calleja Reina, 2025).	RQ1 and RQ3. The artifact is built in Peruvian Spanish and evaluated in one structured therapeutic context in Peru.
Theoretical	There is no design framework that explains how to balance contextual relevance against user agency, privacy, and the risk of homogenizing what children say.	The Stream 1 studies report performance, not design principles, and none describes how the user keeps control over suggestions. Light and McNaughton (2014) define communicative competence but predate generative and context-aware AAC.	RQ1 and RQ3. The thesis derives design principles for human-in-the-loop personalization, following Gregor and Hevner (2013).

Note. RQ1 (design), RQ2 (evaluation), and RQ3 (risk and ethics) are the research questions stated in Deliverable No. 3, Section 05.

The gap that surprised me most. I started this review assuming that there was no AAC research in Spanish for autistic children. The search showed otherwise: EC+ in Spain and EMT en Español

in the United States are recent and peer reviewed. The gap is therefore narrower and easier to defend than the one I had in mind. Intelligent AAC exists only in English, and Spanish-language AAC exists only in conventional form. Nobody has joined the two, and nobody has worked in Latin America. As discussed in Session 4, this is a contextual gap in which the toolkit is ready and only the application is new.

Consequence for the protocol. Griffen et al. (2026) already show that context-aware AAC can work, so the thesis cannot claim novelty for using artificial intelligence. Its contribution is the combination that no included study brings together: Spanish-language localization, multimodal input and output, human-in-the-loop control, and explicit ethical safeguards, evaluated under Design Science Research (Hevner et al., 2004; Peffers et al., 2007). Section 04 of the protocol will be rewritten on this basis.

7. Limitations and Next Steps

- **One human reviewer.** Screening was assisted by an AI tool and checked by me on a sample, but no second person screened the records, so selection bias cannot be ruled out and agreement could not be measured. For the doctoral review, a second reviewer will screen at least a sample.
- **Eligibility judged on abstracts.** The 112 reports were assessed on their abstracts, and the characteristics in Table 6 come from abstracts and bibliographic records. Full texts will be read before data extraction for the thesis. Twenty reports without an abstract were set aside as not retrieved.
- **A cap of ten studies.** Fifty-four eligible reports were left out by the priority rule in Section 3. They include further systematic reviews and many single-case intervention studies, and they will enter the full review.
- **Two sources.** PubMed favors clinical journals and Semantic Scholar is broad but less curated. Scopus, Web of Science, IEEE Xplore, and the ACM Digital Library were not searched, so human-computer interaction work on AAC is probably underrepresented. The strings for Scopus and Web of Science are ready in Section 2.2.
- **Language of the search.** The search terms were in English. Spanish-language studies indexed only in SciELO, Redalyc, or LILACS would not have been retrieved, which matters

because the contextual gap depends on their absence. A Spanish string will be run in those sources before the gap is stated as final.

- **No snowballing and no grey literature.** Backward and forward citation searching from Griffen et al. (2026) and Holyfield et al. (2025) is the next step, together with preprints and conference papers, where much work on artificial intelligence appears first. Several conference papers on AAC with language models were retrieved and removed on title and abstract because they were not journal articles.
- **Not registered.** This is a course exercise and the review was not registered in PROSPERO or OSF. The search strings and the screening log with all 280 records are kept in the course repository.

References

Chavers Edgar, T., Schlosser, R. W., & Koul, R. (2024). Effects of an augmentative and alternative communication intervention package on socio-communicative behaviors between minimally speaking autistic children and their peers. *American Journal of Speech-Language Pathology, 33*(4), 1619–1638. https://doi.org/10.1044/2024_AJSLP-23-00313

Gregor, S., & Hevner, A. R. (2013). Positioning and presenting design science research for maximum impact. *MIS Quarterly, 37*(2), 337–355.

Griffen, B., Raley, S., Holyfield, C., MacNeil, S., Lorah, E. R., & Burns, J. (2026). An examination of context-aware augmentative and alternative communication with automated response options for academic communication from autistic children. *Journal of Special Education Technology*. Advance online publication. https://doi.org/10.1177/01626434261420393

Hevner, A. R., March, S. T., Park, J., & Ram, S. (2004). Design science in information systems research. *MIS Quarterly, 28*(1), 75–105.

Holyfield, C., MacNeil, S., Caldwell, N., O'Neill Zimmerman, T., Lorah, E., Dragut, E., & Vucetic, S. (2024). Leveraging communication partner speech to automate augmented input for children on the autism spectrum who are minimally verbal: Prototype development and preliminary efficacy investigation. *American Journal of Speech-Language Pathology, 33*(3), 1174–1192. https://doi.org/10.1044/2023_AJSLP-23-00224

Holyfield, C., O'Neill Zimmerman, T., MacNeil, S., Sammarco Caldwell, N., Patel, P., Griffen, B., Lorah, E., Dragut, E., & Vucetic, S. (2025). Preliminary investigation of context-aware augmentative and alternative communication with automated just-in-time cloze phrase response options for social participation from children on the autism spectrum. *Folia Phoniatrica et Logopaedica, 77*(3), 269–283. https://doi.org/10.1159/000542304

Light, J., & McNaughton, D. (2014). Communicative competence for individuals who require augmentative and alternative communication: A new definition for a new era of communication? *Augmentative and Alternative Communication, 30*(1), 1–18. https://doi.org/10.3109/07434618.2014.885080

Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., Shamseer, L., Tetzlaff, J. M., Akl, E. A., Brennan, S. E., Chou, R., Glanville, J., Grimshaw, J. M., Hróbjartsson, A., Lalu, M. M., Li, T., Loder, E. W., Mayo-Wilson, E., McDonald, S., . . . Moher, D. (2021). The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *BMJ, 372*, Article n71. https://doi.org/10.1136/bmj.n71

Pak, N. S., Peredo, T. N., Madero Ucero, A. P., & Kaiser, A. P. (2025). EMT en Español Para Autismo: A collaborative communication intervention approach and single case design pilot study. *Journal of Autism and Developmental Disorders, 55*(5), 1770–1787. https://doi.org/10.1007/s10803-024-06322-5

Peffers, K., Tuunanen, T., Rothenberger, M. A., & Chatterjee, S. (2007). A design science research methodology for information systems research. *Journal of Management Information Systems, 24*(3), 45–77. https://doi.org/10.2753/MIS0742-1222240302

Pitt, K. M., Ousley, C., Gibbons, C., Martinez, E., Kirkpatrick, C. E., & Bubak, A. (2026). Artificial intelligence for enhancing emotion expression in augmentative and alternative communication systems: A preliminary study including children both with and without autism. *American Journal of Speech-Language Pathology, 35*(4), 1713–1729. https://doi.org/10.1044/2026_AJSLP-25-00469

Privitera, R., Piccolo, A., De Domenico, C., Leonardi, G., Alito, A., Quartarone, A., Cucinotta, F., & Di Cara, M. (2026). Interventions to promote communication in minimally verbal autistic children: A systematic review. *Medical Sciences, 14*(3), Article 388. https://doi.org/10.3390/medsci14030388

Therrien, M. C. S., Whalon, K., Nunes, D., Marti, P., & Sellers, A. (2025). A systematic review of AAC interventions using speech generating devices for autistic preschoolers. *Augmentative and Alternative Communication, 41*(2), 114–128. https://doi.org/10.1080/07434618.2025.2479768

UNESCO. (2021). *Recommendation on the ethics of artificial intelligence*. UNESCO.

White, E. N., Ayres, K. M., Snyder, S. K., Cagliani, R. R., & Ledford, J. R. (2021). Augmentative and alternative communication and speech production for individuals with

ASD: A systematic review. *Journal of Autism and Developmental Disorders, 51*(11), 4199–4212. https://doi.org/10.1007/s10803-021-04868-2

Zurita Díaz, A. J., & Calleja Reina, M. (2025). The impact of EC+ as a multimodal support in intervention for children with autism spectrum disorder: A technological alternative. *Disability and Rehabilitation: Assistive Technology, 20*(8), 3027–3043. https://doi.org/10.1080/17483107.2025.2520881

Appendix A. Access Links to the Included Studies

Table A1

DOI and PubMed Links for the Ten Included Studies

ID	Citation	Links
A1	White et al. (2021)	https://doi.org/10.1007/s10803-021-04868-2 https://pubmed.ncbi.nlm.nih.gov/33511525/
A2	Chavers Edgar et al. (2024)	https://doi.org/10.1044/2024_AJSLP-23-00313 https://pubmed.ncbi.nlm.nih.gov/38771825/
A3	Holyfield et al. (2024)	https://doi.org/10.1044/2023_AJSLP-23-00224 https://pubmed.ncbi.nlm.nih.gov/38290536/
A4	Holyfield et al. (2025)	https://doi.org/10.1159/000542304 https://pubmed.ncbi.nlm.nih.gov/39467532/
A5	Therrien et al. (2025)	https://doi.org/10.1080/07434618.2025.2479768 https://pubmed.ncbi.nlm.nih.gov/40164143/
A6	Pak et al. (2025)	https://doi.org/10.1007/s10803-024-06322-5 https://pubmed.ncbi.nlm.nih.gov/38613594/
A7	Zurita Díaz and Calleja Reina (2025)	https://doi.org/10.1080/17483107.2025.2520881 https://pubmed.ncbi.nlm.nih.gov/40526487/
A8	Griffen et al. (2026)	https://doi.org/10.1177/01626434261420393 https://pubmed.ncbi.nlm.nih.gov/42454091/
A9	Pitt et al. (2026)	https://doi.org/10.1044/2026_AJSLP-25-00469 https://pubmed.ncbi.nlm.nih.gov/42257585/
A10	Privitera et al. (2026)	https://doi.org/10.3390/medsci14030388 https://pubmed.ncbi.nlm.nih.gov/42506357/

Note. The DOI is the permanent link to the publisher's version. The PubMed link opens the indexed record and abstract.

Appendix B. PRISMA 2020 Items Covered in This Deliverable

Table B1

Location of the PRISMA 2020 Checklist Items That Apply to This Mini Review

Item	Checklist topic	Where it is reported
1	Title identifies the report as a systematic review	Cover and title
3, 4	Rationale and objectives	Section 1
5	Eligibility criteria	Section 3, Table 4
6	Information sources and dates	Section 2.3, Table 3
7	Full search strategy for each source	Section 2.2
8	Selection process	Section 3
9, 10	Data collection and data items	Section 3
11	Risk of bias assessment	Section 3 and Section 4 (appraisal)
16a	Flow of records through the review	Section 5, Figure 1, Tables 7 and 8
16b	Studies that met the criteria but were excluded	Section 3 (limit of ten studies) and screening log
17	Characteristics of included studies	Section 4, Tables 5 and 6
23	Interpretation, limitations, implications	Sections 6 and 7
24	Registration and protocol	Section 7
27	Availability of data and materials	Section 7

Note. Items on meta-analysis, effect measures, and certainty of evidence (12 to 15 and 19 to 22) do not apply because no statistical synthesis was carried out.